Endometriosis On Social Networks

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Introduction

Importance and relevance of the problem

Brilliant!

Endometriosis affects roughly 10% of reproductive-age women worldwide (Zondervan et al. 2018), yet diagnosis is often delayed for many years due to social stigma and the normalization of menstrual pain by both patients and physicians. The condition is not only biomedical but also a social and informational phenomenon revealing structural inequalities in health awareness. Recent social media activism has increased visibility, reduced stigmatization, and encouraged earlier care-seeking. From a data-science and complexity perspective, the spread of awareness follows small-world diffusion patterns, where information and empathy travel through densely connected local clusters. Thus, recognizing endometriosis is also a spatial-network problem shaped by social position, urban context, and online community structure.

citation

State of research and approaches

diffusion across commun. is difficult

this is more so in the case of stigma

While endometriosis-specific network studies are rare, related healthcare and social-network research can serve as a conceptual starting point. Small-world networks are characterized by short average path lengths and high clustering, which enable rapid diffusion of information and behavioral patterns (Watts and Strogatz 1998). This logic has been extended to economic and urban systems, where dense local structures and a few long-range connections facilitate innovation and knowledge flows (Fleming, King III, and Juda 2007). Within the framework of economic complexity, countries and industries grow by combining and reusing capabilities that are linked through networks of related products (Hidalgo and Hausmann 2009). Similar network-based mechanisms appear in health research, where spatial and social connections shape the diffusion of information and health outcomes (Rosenquist, Fowler, and Christakis 2011). In particular, recent findings show that antidepressant use is strongly correlated with spatial social network structures (Lengyel et al. 2024), highlighting how network position affects wellbeing.

I think these are different!

would be better to cite diff. papers *

Research niche and question

Many studies have examined small-world and urban networks and their effects on health (Bailey et al. 2018). Other research has focused on how endometriosis appears in social media (Adler et al. 2024) and on the broader links between the economy and healthcare (McClellan, McNeil, and Newhouse 1994). However, these areas are usually studied separately. We still know little about how social networks—both online and offline—shape the spread of knowledge, the reduction of

prosperity

* Chen... et al. 2017 NComms

* Watts twitter diffusion paper

stigma, and the spatial pattern of diagnosis in chronic diseases such as endometriosis. Online communities and influencers may help information travel faster and make the condition more visible, but there is little empirical evidence for this* (Aral 2016). This study explores how the structure and dynamics of social networks shape the recognition and treatment of endometriosis. It examines whether higher network density, strong central actors, and greater online visibility lead to faster diffusion of awareness, lower stigma, and earlier diagnosis across social and spatial contexts.

What is the role of doctors / famous patients? → Dr. Fülöp Zita Patai

Stakeholders and potential benefits

awareness is also a diff. among doctors — diagnosis is usually very slow ⇓ can AI help?

The main stakeholders are women affected by endometriosis and the healthcare professionals who diagnose and treat them. Greater public awareness can help patients recognize symptoms earlier and reduce the frustration that often arises from delayed or dismissive medical responses. For healthcare workers, understanding how information and attitudes spread through social networks may improve communication and diagnostic sensitivity (Rosenquist, Fowler, and Christakis 2011). On a social level, the formation of smaller support clusters and online communities allows women to connect, share experiences, and feel less isolated (Bailey et al. 2018). This can reduce stress—a known factor that worsens symptoms—and ultimately improve both wellbeing and healthcare outcomes.

Data Collection

This project builds upon the conceptual and methodological foundation of (Lengyel et al. 2024), whose study linked individual-level antidepressant use with the structural properties of a nationwide online social network. While their approach relied on full-population iWiW friendship data and precise medical records, the present study adapts this framework to the context of endometriosis. Instead of individual-level health information, I rely on aggregated epidemiological indicators of endometriosis burden, and instead of a full-scale social network, I construct a network from publicly available social media interactions. The underlying logic, however, remains similar: the aim is to examine whether the structure of online relational ties helps explain spatial and temporal patterns in health-related awareness.

The health outcome in this project is defined through *aggregate measures of endometriosis burden* (Zondervan et al. 2018). These may include incidence (new cases per 100 000 women), prevalence, hospital discharge statistics, or age-standardised estimates from the Global Burden of Disease (GBD) database at the country or regional level. Such measures serve as a realistic epidemiological baseline that can be meaningfully compared with social media activity. By using aggregated instead of individual-level data, the study avoids privacy concerns while preserving the ability to describe diagnosis patterns at a population level. Where?

The network data are derived from social media posts, primarily from X/Twitter. Endometriosislets do this! :)

related content is identified using widely used hashtags such as #endometriosis, #endowarrior, and #endometriosisawareness. From each post, I extract the user identifier, timestamp, textual content, self-reported location (used to assign posts to a country or region), and the presence of user-to-user interactions such as replies, mentions, or retweets. On the basis of data availability, a single pilot location (country or subnational region) is selected, ensuring both sufficient social media activity and reliable epidemiological data. Monthly temporal resolution is used wherever possible, and the examined period is chosen to include intervals of heightened public attention, such as Endometriosis Awareness Month or major media events.

Cool!

To complement the social media corpus, I collect Google Trends time series for the search term "endometriosis". These data serve as a proxy for public awareness, reflecting broader information-seeking behaviour. Google Trends provides at least country-level geographical resolution, and occasionally regional resolution depending on the location. By combining social media content, user-interaction data, and Google search interest, the dataset captures multiple dimensions of awareness dynamics.

Methodology

Network Construction

A user-interaction network is constructed where nodes represent social media users who posted about endometriosis during the selected period, and edges represent directed or undirected interaction ties. Directed edges arise from the structure of the platform (reply $A \rightarrow B$, mention $A \rightarrow B$, retweet $A \rightarrow B$), while an undirected projection may be used for certain analyses where directionality is not essential. The resulting graph captures the relational environment through which awareness can spread.

Two key structural measures are computed in analogy with the *local cohesion* (LC) and *spatial diversity* (SD) variables of Lengyel et al. (2024). These measures quantify the extent to which a location's online discussion is internally cohesive and externally connected. ~~Explain what you expect from these measures.~~

Bonding Structure (LC Analogue)

Let c denote the selected location (country or region). I first extract the subgraph consisting of users whose self-reported location corresponds to c. The *local cohesion* of this subgraph is quantified using the clustering coefficient:

$$LC_c = \frac{3T_c}{W_c},$$

where T_c is the number of triangles and W_c the number of connected triples in the local network. High values of LC_c indicate dense, tightly knit online communities, characterised by extensive triadic closure and strong bonding social capital (Coleman 1988). Such cohesion suggests that information may circulate efficiently within the local cluster.

Bridging Structure (SD Analogue)

To capture bridging social capital, I quantify the geographical diversity of interaction ties originating from location c. Let p_i denote the proportion of interactions from users in c to users located in region or country i. The *spatial diversity* measure is defined using Shannon entropy:

$$SD_c = -\sum_i p_i \ln(p_i).$$

A high value of SD_c indicates that users in c interact with a wide range of other geographical locations, forming long-range connections that may facilitate the early or rapid inflow of information, including medical knowledge or personal experiences related to endometriosis.

Awareness and Epidemiological Indicators

For each location c and time period t (typically months), I construct the following variables:

- $GT_{c,t}$: Google Trends index for the query "endometriosis";
- $SM_{c,t}$: number of endometriosis-related social media posts per 100 000 inhabitants;
- $Inc_{c,t}$: incidence or another epidemiological indicator of endometriosis burden.

These measures allow for the joint examination of online attention, social network structure, and health-related outcomes.

Analytical Framework

The first analytical step is to model awareness as a function of social media dynamics and network structure. A simple linear specification takes the form

$$Awareness_{c,t} = \alpha + \beta_1 LC_c + \beta_2 SD_c + \beta_3 SM_{c,t} + \varepsilon_{c,t},$$

the networks can be also dynamic

where awareness may be operationalised by Google Trends or by the intensity of local social media engagement.

To explore the association between awareness patterns and epidemiological outcomes, I estimate:

$$Inc_{c,t} = \gamma + \delta_1 LC_c + \delta_2 SD_c + \delta_3 Awareness_{c,t-1} + u_{c,t},$$

where the lag in awareness reflects the assumption that information diffusion precedes diagnosis behaviour. Since the project is a pilot study, the objective is not causal identification but the detection of meaningful associations between network structure, awareness, and recorded endometriosis burden.

Expected Outcomes

I expect that locations exhibiting stronger bridging structures (higher SD_c) and more cohesive internal discussion networks (higher LC_c) will display earlier and more intense awareness spikes (Aral 2016). These dynamics should be visible both in rising Google Trends search interest and in increased volumes of social media activity. A further expectation is that such awareness surges may be followed by increases in the recorded number of endometriosis diagnoses, consistent with prior findings showing that network structures can shape health-related behaviours and outcomes (Rosenquist, Fowler, and Christakis 2011). Importantly, this does not imply that the disease becomes more common; rather, heightened awareness may lead previously undiagnosed individuals to recognise symptoms and seek medical care.

In particular, regions with more diverse external connections may receive information from multiple sources simultaneously, accelerating both the spread of symptom recognition and the perceived legitimacy of the condition. Likewise, highly cohesive local clusters may reinforce shared experiences and encourage individuals to reinterpret previously normalised symptoms. If these mechanisms hold, then changes in network structure should predict not only short-term fluctuations in online attention, but also medium-term shifts in help-seeking behaviour. Ultimately, the model anticipates that awareness diffusion is a precursor to measurable improvements in diagnostic rates, especially in areas where endometriosis has historically been under-recognised.

Impact

Endometriosis is known for long diagnostic delays and widespread underdiagnosis, partly due to social stigma and limited public understanding of the condition. By analysing how awareness spreads across online social networks, this project contributes to understanding the social mechanisms that influence diagnosis patterns. The findings may help identify the types of network structures and key actors (including potential bridging nodes or influencers) that accelerate the diffusion of reliable information. Insights from this study may inform public health communication, guiding awareness campaigns toward regions where diffusion is likely to be slower, ultimately reducing undiagnosed suffering and improving health outcomes for individuals affected by endometriosis.

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